

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER ARCHITECTURE COURSE CODE: CSA12

COURSE OUTCOME COVERED

CO1: Components of a computer system, including CPU, memory, and I/O. Basic Operational Concepts: Instruction execution cycle, instruction fetch and decode. Bus Structures: Single-bus, multi-bus, and hierarchical buses. Factors of Performance: CPU execution time, CPI, clock cycles, and benchmarking. 8085 and 8086 Architectures: Overview, instruction sets, and addressing modes. Modern Architectures: ARM Cortex, Intel Core, and AMD Ryzen. Instruction Sets, Formats, and Addressing Modes. Number Systems: Binary, hexadecimal, and their applications in computer systems. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks:25

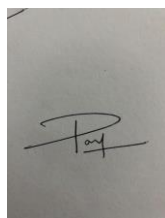
ANNEXURE A CONCEPT MAPPING

Code of Conduct:

I PAVAN.P/ 192411017 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with a handwritten signature in black ink. The signature appears to be 'Pavan.P.' with a stylized flourish.

ANNEXURE A

1. Construct a concept map that connects the components of a computer system (CPU, memory, and I/O) with the instruction execution cycle. Clearly show how instruction fetch, decode, and execution stages interact with memory and input/output operations during program execution.
2. Develop a concept map linking different bus structures (single-bus, multi-bus, hierarchical bus) with system performance factors such as CPU execution time, clock cycles, and CPI. Illustrate how bus design influences data transfer speed and overall system efficiency.
3. Create a concept map that compares 8085 and 8086 architectures with modern architectures like ARM Cortex, Intel Core, and AMD Ryzen. Show relationships in terms of instruction sets, addressing modes, processing capability, and architectural advancements.
4. Design a concept map that integrates instruction sets, instruction formats, and addressing modes with number systems (binary and hexadecimal). Explain how instructions are represented, decoded, and executed using these number systems in a computer.
5. Prepare a concept map that connects benchmarking techniques with performance metrics such as CPU execution time, CPI, and clock cycles, and relate these to different architectures (8085, 8086, and modern processors). Demonstrate how performance evaluation varies across different system designs.

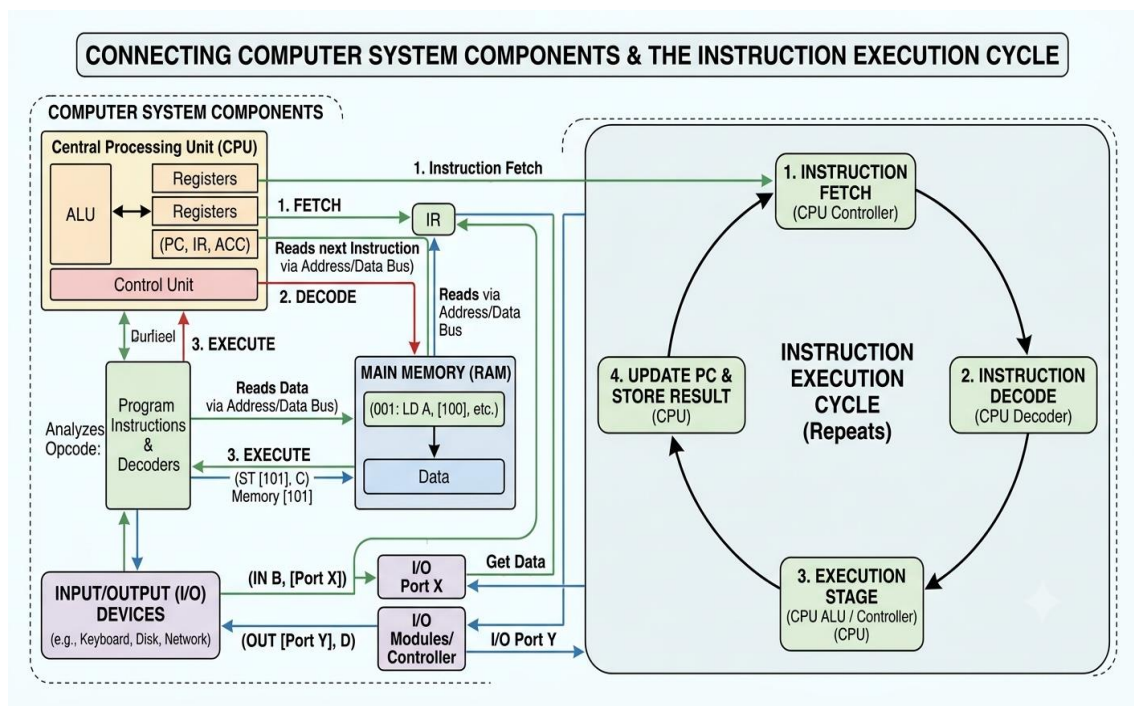
MARKS ALLOCATION

Criteria	Understanding of Problem Context (20%)	Application of Concepts (20%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (20%)
Q1					
Q2					
Q3					
Q4					
Q5					

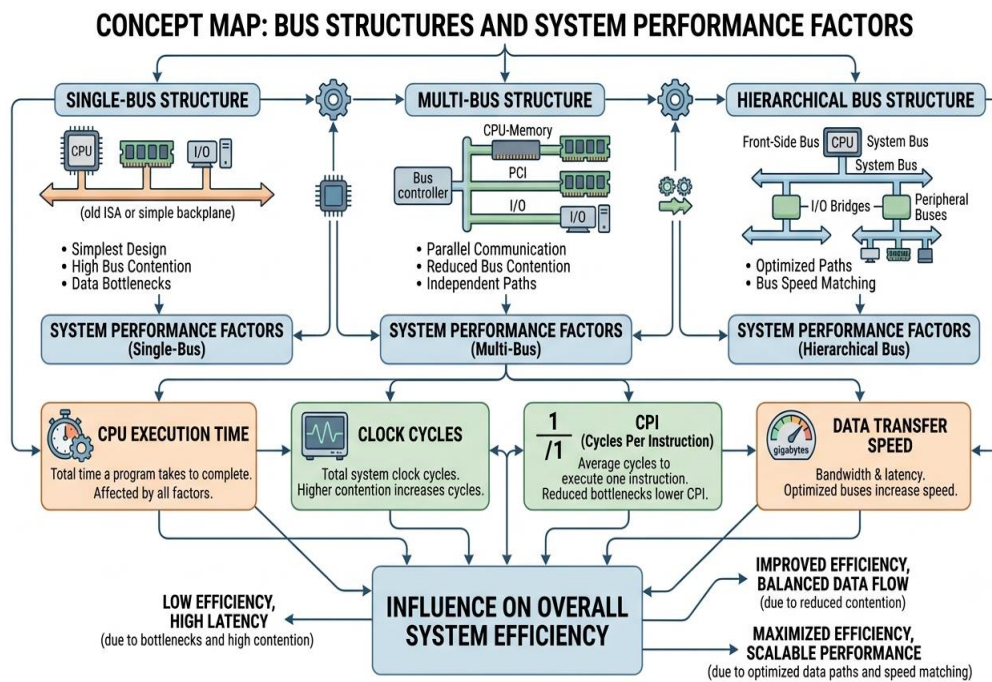
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1. ANSWER: -



2. ANSWER: -

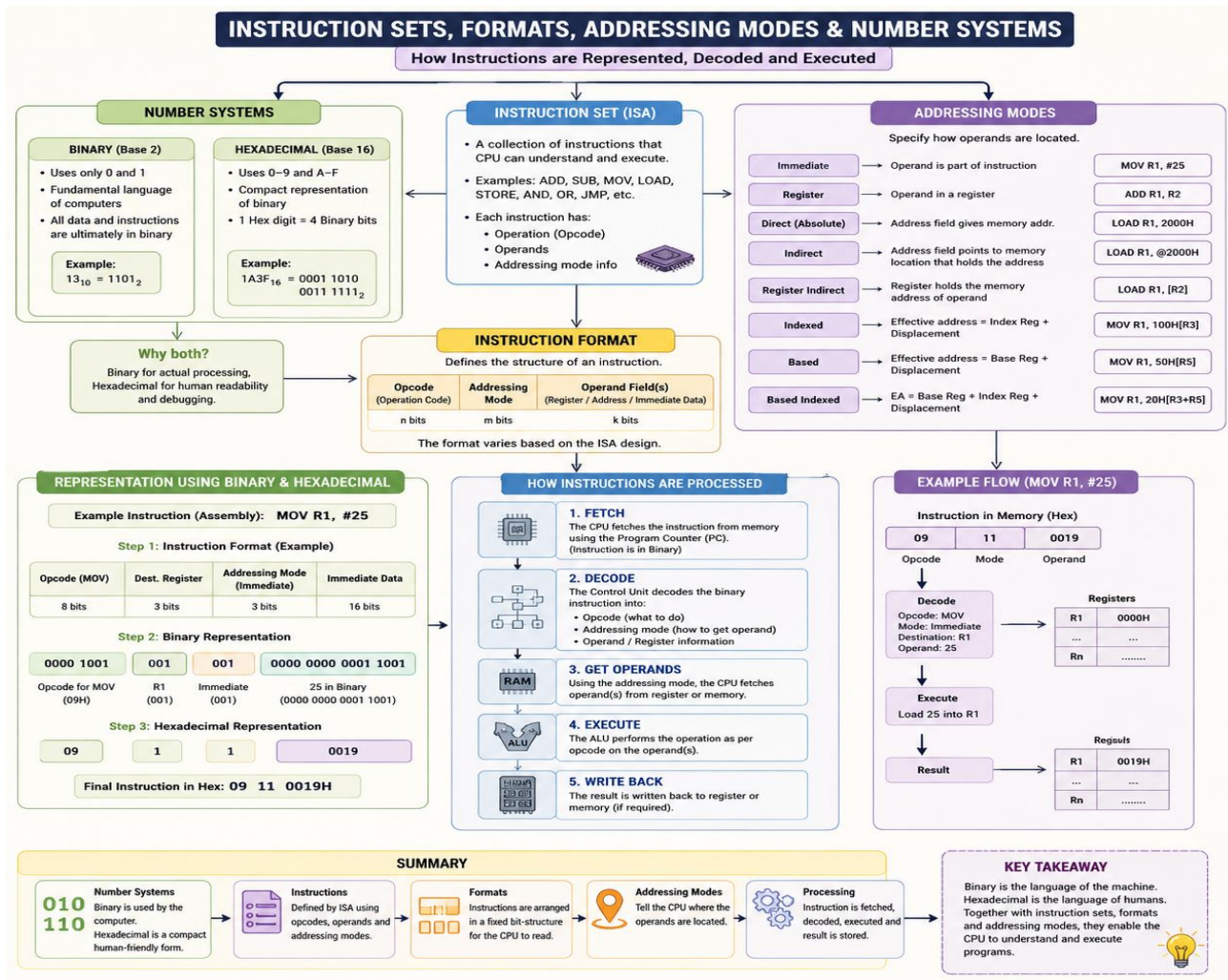


3. ANSWER: -

Comparison of Microprocessor Architectures

Feature	8085 (8-bit)	8086 (16-bit)	Intel Core / AMD Ryzen (x86-64)	ARM Cortex (RISC)
Bit Architecture	8-bit	16-bit	64-bit	64-bit
Architecture Type	Accumulator-based (CISC)	General-purpose register (CISC)	Advanced Superscalar (CISC)	Load/Store Architecture (RISC)
Memory Limit	64 KB	1 MB (with segmentation)	Terabytes (Very high)	Terabytes (Very high)
Register Set	Basic (A, B-E, H-L, H-L, SP, PC, flags)	Enhanced (AX-DX, SI, DI, BP, SP, IP, flags)	Extensive (RAX-RDX, R15, multi-level cache, etc.)	General-purpose (X0-X30), specialized extensions
Instruction Set	Very simple 8-bit	More powerful 16-bit	Extremely complex, dynamic execution, SIMD	Simple, mostly fixed-length, optimized
Addressing Modes	Direct, Register, Immediate	Complex (Base, Index, Segment offset)	Very complex, dynamic translation	Very simple, optimized
Key Advancements	Hardwired control, early standard	Microcoded control, major evolution	Multicore, dynamic parallelism, out-of-order execution, AVX	Efficient pipelining, security, SIMD, scalable

4. Answer: -



5. ANSWER: -

CONCEPT MAP: BENCHMARKING TECHNIQUES, PERFORMANCE METRICS AND ARCHITECTURE COMPARISON

